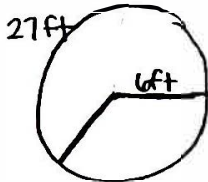


For ALL of the following problems you must a) draw a picture, and b) solve the problem. Be sure to write your answer with the units represented.

- 1) Find the radian measure of the central angle of a circle with a radius of 6 feet and an intercepted arc of 27 feet.

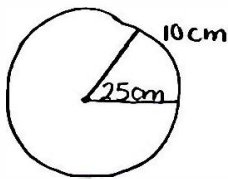


$$\theta = \frac{s}{r}$$

$$\theta = \frac{27ft}{6ft}$$

$$\theta = 4.5 \text{ radians}$$

- 2) Find the radian measure of the central angle of a circle with a radius of 25 centimeters and an intercepted arc of 10 centimeters.



$$\theta = \frac{10cm}{25cm}$$

$$\theta = 0.4 \text{ radians}$$

For the following questions find the length of an arc that is intercepted by the given central angle in a circle with the given measure.

- 3) central angle of 120° and radius of 9 miles.

$$s = \theta r$$

$$\frac{120^\circ}{1} \cdot \frac{\pi}{180} = \frac{2\pi}{3} \text{ rad}$$

$$s = \frac{2\pi}{3}(9)$$

$$s = 6\pi \text{ mi}$$

- 5) central angle of $\frac{11\pi}{6}$ and radius of 18 millimeters.

$$s = \frac{11\pi}{6}(18)$$

$$s = 33\pi \text{ mm}$$

- 4) central angle of 50° and radius of 3 centimeters.

$$s = \frac{5\pi}{18}(3)$$

$$\frac{50^\circ}{1} \cdot \frac{\pi}{180} = \frac{5\pi}{18} \text{ rad}$$

$$s = \frac{5\pi}{6} \text{ cm}$$

- 6) central angle of $\frac{3\pi}{4}$ and diameter of 10 feet.

$$s = \frac{3\pi}{4}(5)$$

$$s = \frac{15\pi}{4} \text{ ft}$$

7.) Find the angular and linear velocity (speed) of a gear with radius of 10 centimeters if one revolution takes 14 seconds.

$$2\pi$$

$$A.S. = \frac{2\pi}{14} \text{ rad/sec}$$

$$L.S. = \left(\frac{\pi}{7} \text{ rad/sec}\right)(10 \text{ cm})$$

$$A.S. = \frac{\pi}{7} \text{ rad/sec.}$$

$$L.S. = \frac{10\pi}{7} \text{ cm/sec}$$

8.) Find the angular and linear velocity (speed) of a wheel with radius of 35 inches if half a revolution takes 70 seconds.

$$\pi$$

$$A.S. = \frac{\pi}{70} \text{ rad/sec.}$$

$$L.S. = \frac{\pi}{70}(35)$$

$$L.S. = \frac{\pi}{2} \text{ in/sec.}$$

9.) Find the angular and linear velocity (speed) of a gear with a diameter of 4 meters if it makes 5 revolutions per minute.

$$r = 2 \text{ m}$$

$$10\pi$$

$$A.S. = \frac{10\pi}{1}$$

$$L.S. = 10\pi(2)$$

$$A.S. = 10\pi \text{ rad./min}$$

$$L.S. = 20\pi \text{ m/sec.}$$

10) Find the angular and linear velocity (speed) of a wheel with radius 120 feet if it rotates 240° in 30 seconds.

$$A.S. = \frac{4\pi}{3} \left(\frac{1}{30}\right)$$

$$L.S. = \frac{\pi}{45}(120)$$

$$\frac{240}{1} \frac{\pi}{180} = \frac{4\pi}{3}$$

$$A.S. = \frac{\pi}{45} \text{ rad/sec.}$$

$$L.S. = \frac{8\pi}{3} \text{ ft/sec.}$$

11.) Kelly, Andrew, Cheri, and Hayden are headed to Florida for Spring Break. They are traveling in a car that has 24 inch tires. How fast are they traveling if 2 revolutions take 1.5 seconds? Express your answer in miles per hour. Give both the exact answer and the answer rounded to 1 d.p.



$$2 \text{ rev} = 4\pi$$

$$t = 1.5 \text{ sec.}$$

$$A.S. = \frac{4\pi}{\frac{3}{2}} \text{ rad/sec}$$

$$A.S. = 4\pi \left(\frac{2}{3}\right) \text{ rad./sec.}$$

$$A.S. = \frac{8\pi}{3} \text{ rad/sec.}$$

$$L.S. = \frac{8\pi}{3}(12)$$

$$L.S. = 32\pi \text{ in/sec}$$

$$L.S. \approx 100.531 \text{ in/sec}$$

$$\frac{32\pi \text{ in}}{1 \text{ sec}} \cdot \frac{1 \text{ ft}}{12 \text{ in}} \cdot \frac{1 \text{ mi}}{5280 \text{ ft}} \cdot \frac{3600 \text{ sec}}{1 \text{ hr}}$$

$$L.S. \approx 5.712 \text{ mi/hr.}$$